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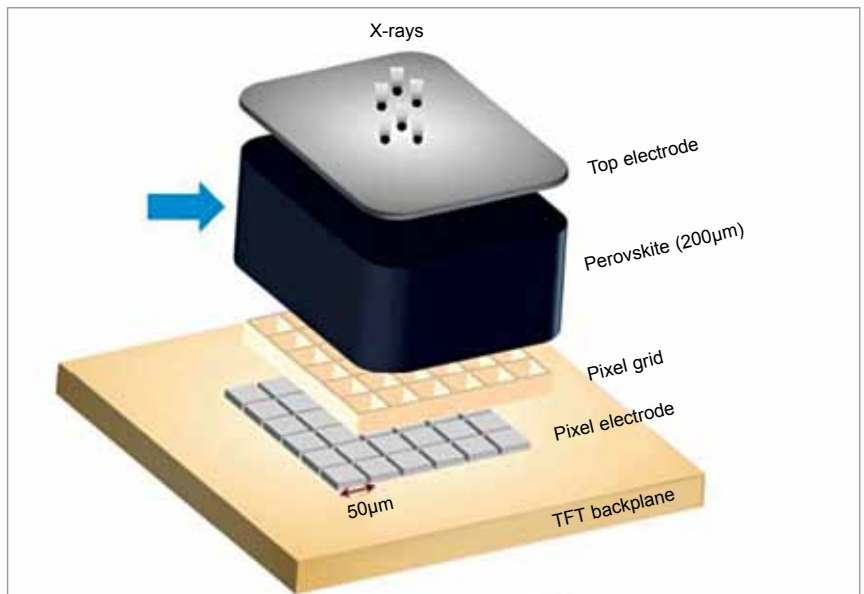


Fig. 9: Perovskite based X-ray sensor (Credit: Holst Center)

here, scenes of no change will be ignored.

This is highly beneficial for machine vision applications as seen from the comparison in Fig. 8.

Only changing subjects such as moving vehicles and people relative to immovable backgrounds such as buildings get recorded.

Other advantages include:

- Higher dynamic range
- Greater temporal resolution
- Less data production, making data transmission much easier
- Simple data processing (as static image areas get ignored by the hardware)

However, the technology has some drawbacks such as difficulty in interpreting datasets without specialist software, greyscale image, and less extensive analysis algorithms than conventional image analysis.

Applications include:

- Autonomous vehicles
- Drones
- Occupant tracking in smart buildings
- Iris-tracking in AR/VR goggles

Flexible X-ray sensors

Conventional X-ray sensing works when the X-ray wavelengths reflected from an object hit a digital X-ray image detector. This image detector consists of a layer known as a scintillator that converts the X-rays into visible light,

which is further refined by a silicon photodetector. The emerging amorphous silicon X-ray detector takes this one step ahead by allowing devices to become flexible and lightweight enough to be wrapped around curved body parts.

Another exciting innovation in this field is the perovskite based X-ray sensor in which a very thin (200µm) layer of perovskite absorbs the X-rays and creates charges. The sensor is beneficial for generating high-resolution images.

The road ahead

Most of the sensors discussed here are suited for machine vision and image analysis across many applications from cameras to ADAS and drones. These will also serve as a cheaper alternative to current InGaAs sensors. While some of them are nearing the end of the development phase, others are still in it. Nevertheless, most of them would be rolled out into the markets in the next few years.

Hyperspectral imaging based sensors are quite established. Among the SWIRs, the extended silicon is going to be the first one to arrive in markets by 2022. Event based imaging is expected to follow a year after. **EFY**

This article is compiled by Vinay Prabhakar Minj, a technology journalist at EFY and is based on an IDTechEx webinar